

BATSAMBUU BATBOLD

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EDUCATION

Macalester College | St. Paul, MN | Expected May 2027

B.A. Mathematics & Computer Science | GPA: **3.96/4.0**

Relevant Coursework: Probability, Network Science, Statistical Machine Learning, Computational Linear Algebra, Database Management Systems

United World College Maastricht | Netherlands | May 2023

IB Diploma (43/45), Bilingual Diploma

HONORS & AWARDS

- **Bronze Medalist, International Mathematical Olympiad** (2021): Placed top 250 globally in the World Championship Mathematics Competition for High School students
- **1st Place, Macalester ASA DataFest** (2025): Exploratory data analysis on large-scale geospatial datasets in R; presented actionable location recommendations to a panel of judges.
- **1st Place, Konhauser Problemfest** (2026): First Macalester victory since 2018 across a 5-college competition (Macalester, Carleton, St. Olaf, St. Thomas, Augsburg).
- **2× 1st Place, Macalester Local Programming Contest** (2025, 2026): Competitive programming problems in C++.

EXPERIENCE

Mathematics Domain Expert (Frontier AI) | Mercor (Remote) | Dec. 2025 – Jan. 2026

- Authored competition-level math problems to evaluate state-of-the-art AI models; assessed model-generated proofs for logical correctness and provided annotated solutions for AI training data.

Full Stack Developer & Organizer | Young Mathematicians of Khuvsigul NGO | May 2019 – Present

- Built a Mongolian-language [olympiad training platform](#) with problem archives, courses, and live competitions; wrote SQL queries with indexing for search and data retrieval.
- Grew platform to 100+ registered users and 1,300+ daily visitors; organize free math camps and weekly online contests for 30+ rural students.

Undergraduate Preceptor | Macalester College | Jan. 2024 – Present

- Support 30+ students per semester in Computational Linear Algebra and Probability; hold office hours and grade assignments.

Teaching Assistant | Delgermurun Secondary School | Summer 2025

- Coached students for the International Math Contest for Juniors; one student earned a Silver Medal. Designed a Quarto [website for math and programming lessons](#).

RESEARCH & PUBLICATION

Research Assistant & Data Analyst | Macalester College | Summer 2026

- Analyzed a network of scientific concepts from 70 million academic papers across 150+ fields using Topological Data Analysis (TDA) to identify knowledge gaps and concept clusters.
- Built [TopoLens](#), an interactive persistent homology visualizer enabling researchers to detect structural holes in knowledge networks and generate publication-ready data visualizations.
- Developed custom graph-layout algorithms for 10,000+ node networks, reducing cycle-edge crossings by ~15%.

From Chaos to Continents | Python, NumPy, Streamlit, Blender

- Engineered a procedural terrain generation engine using Voronoi Diagrams and Perlin Noise to create 50,000+ polygon worlds.
- Accepted and presented at The International Symposium on Computational Geometry (SoCG 2026) as a multimedia submission.

PROJECTS

Art2Vec: Quantitative Analysis of Art Movements | Python, Pandas, Scikit-learn, SciPy, NetworkX | May 2026

- Modeled 16,000+ paintings as co-occurrence networks using Louvain community detection and centrality metrics.
- Built a content-based recommendation system using feature vectors, cosine similarity, and k-nearest neighbors.

NBA 2K Player Rating Simulator | Python, Pandas, Scikit-learn, Streamlit | Dec. 2025

- Trained XGBoost and Random Forest regression models to predict NBA 2K ratings from real-world stats, achieving an average MAE of 8.58 across 36 attributes, 5-fold cross-validated.

TECHNICAL SKILLS

Technical Areas: Quantitative Research, Statistical Modeling, Machine Learning, Topological Data Analysis, Network Analysis, Regression Analysis, Data Visualization, SQL Query Optimization

Languages: Python, R, SQL, C++, Java, TypeScript, JavaScript

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, SciPy, NetworkX, Matplotlib, Streamlit, React

Tools: Git, PostgreSQL, LaTeX, Jupyter Notebooks